1. $\underbrace{\bar{R} \otimes R \otimes R_A}_{(T_R^a)^m{}_n} = 1 \oplus \cdots$

$\bar{R} \otimes R = \underbrace{1}_{\delta^m{}_n} \oplus R_A \oplus \cdots$

$SU(N)$ R_F $\bar{R}_F \otimes R_F = 1 \oplus R_A$

$\underbrace{x^{ij} \otimes \psi_i}_{N^2} = 1 + N^2 - 1 = N^2$

$SO(N)$ R_F: $\bar{R}_F = R_F$. $R_F \otimes R_F = 1 \oplus R_A \oplus R_S$

$T(R_A) = 2N - 4$

2. 反常系数. G R T_R^a

$A(R) d^{abc} = \frac{1}{2} tr[T_R^a \underbrace{\{T_R^b, T_R^c\}}]$

$\downarrow$

$T_R^b T_R^c + T_R^c T_R^b$

最低维复表示: $A(R) = 1$.

实/赝实: R $A(R) = 0$.

3. G 连通非紧单群. 无非平庸有限维幺正表示.

$SO(1,3)$. 如 boost 变换非幺正.

4. $\Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu g_{\rho\sigma} = g_{\mu\nu}$, $\Lambda^\rho{}_\mu \approx \delta^\rho{}_\mu + \delta\omega^\rho{}_\mu$

$\Rightarrow \quad \delta\omega^{\mu\nu} = -\delta\omega^{\nu\mu}$ 6.